

Networking Issues in the Azure Landing Zone

Problem	Solution or Reason
No connection to the CNDTAG possible from Azure Subscription? Why can I not reach my server in Azure?	The resources in the Azure Landing Zone need to be inside an Enterprise Subscription (cid or cip Subscription) and inside the vnet_cid/p_xxxx virtual network. Inside that the resource must be mounted in the 10.206.xx.xx range and not a 100.xxx.xxx range, as this one is the only range where the on-premise CNDTAG is routed. After that the nsg-defaultnetworking needs to be configured properly to be able to reach the service out- or inbound. The NSG (Network Security Group) acts as an Azure internal firewall. The Pannet Firewall is another firewall that is between the Azure Landing Zone and the on-premise CNDTAG. This one also needs to be cleared by opening a ticket to the landing zone team (link1) with an excel sheet that includes every rule (not included rules in the sheets will be removed from the firewall). A new template excel sheet can be downloaded here (link2). The Tufin Firewall, which is the internal on-premise Firewall also needs to be cleared in order to reach your services, even though the Pannet Firewall is already cleared. Please refer to the documentation of the Tufin FW. In some cases, the blocking instance regarding network issues is the firewall on the Virtual Machine itself (e.g. IP-Tables, Windows Firewall Settings etc.)
What can be reached from Azure in the CNDTAG?	The reachable range is 10.0.0.0/8. No other range can currently be reached from Azure.
Why can't I not reach my service from the CNDTAG (DNS entry not resolvable). Why is my address not resolvable?	The current DNS solution requires adding an A (alias) entry into the subscription owned private DNS zone (named like the subscription). To reach it, an additional cname entry needs to be added to a central private DNS zone (depending on the service there are different central DNS zones) by the Azure Team. This can be done via a ticket here (link3)
The connection inside my virtual network is not working? The connection between my resources is not working?	The connection between resources inside a virtual network regardless in which address space they are deployed, must be cleared in the nsg-defaultnetworking. The nsg-defaultnetworking is always securing your Subnets in the VNET.
Can I add a custom NSG on my resources that are connected to CNDTAG	No, this is not possible, as the nsg-defaultnetworking is mandatory to ensure a secure connection.
How to protect private endpoints? My private endpoint can be reached from everywhere – network security groups are not working. How can I solve this problem?	Private endpoints must be secured with Application Security Groups. If you don't do that – especially in the Azure Portal, any network security group (NSG) on the subnet will not work for private endpoints. That means, private endpoints can be reached from all connected networks (whole cndTAG). In case that happened, you have to enable this policy again, before Application security groups works again.

	Code example: <pre>''' az network vnet subnet update \ --disable-private-endpoint-network-policies false \ --name default \ --resource-group myResourceGroup \ --vnet-name myVNet '''</pre>
How should I make tier separation in Azure?	Always use application security groups for it. Don't use additional network security groups on network interfaces except ASGs